

SEFIRE

THE SHARED EXPENSE TRACKER

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ELEVATOR PITCH

SEFire is a shared expense tracker designed for roommates, couples, and households. It allows users to record shared expenses, customize how costs are split, track outstanding balances, and record payments in one place. SEFire helps users better understand their finances by providing budgeting guidance, financial vocabulary tooltips and access to custom variable expenses. These features help make managing shared household expenses more organized and easier to for everyone.

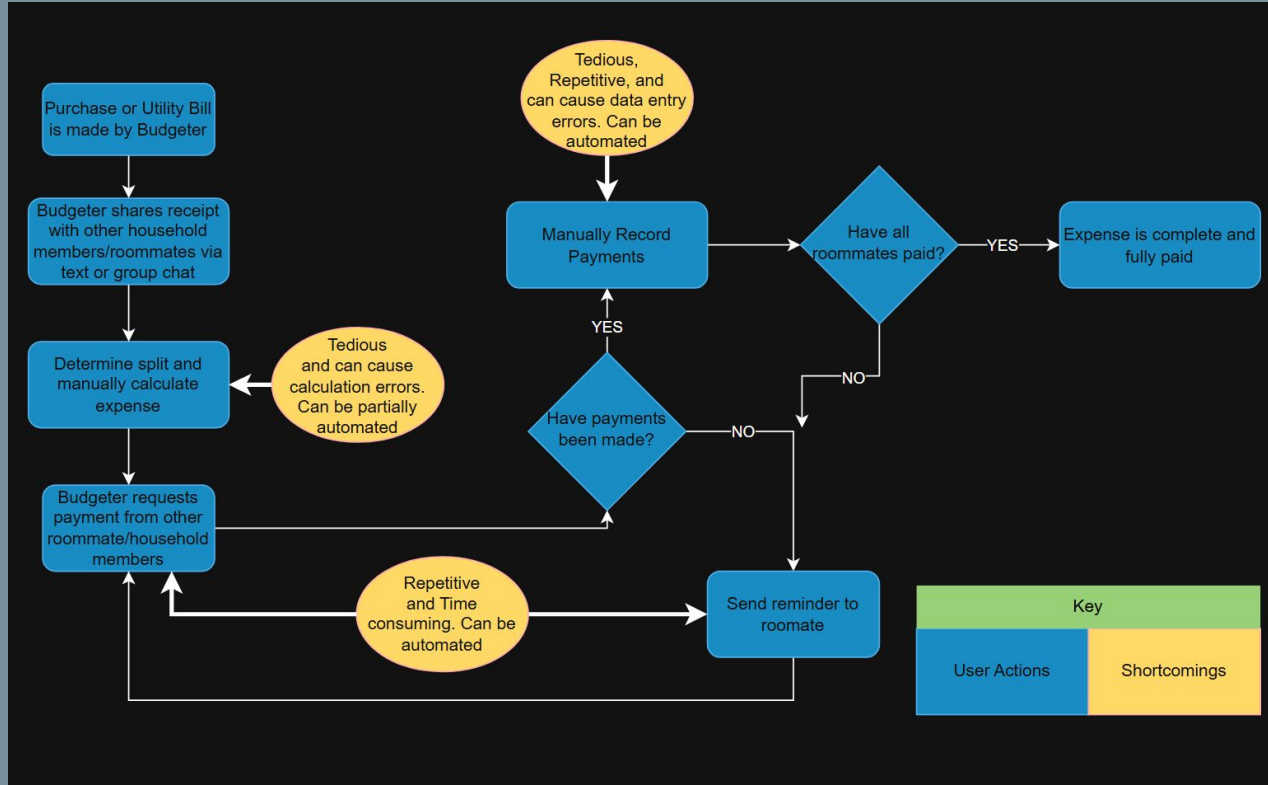
PROBLEM STATEMENT

Managing financial responsibilities in shared households can be difficult. Tracking who pays for rent, utilities, groceries, and other shared expenses can be confusing and may lead to missed payments or disagreements.

PROBLEM CHARACTERISTICS

- Missed payments
- Conflict between household members/roommates
- Confusion with processing multiple expenses
- Difficulty in tracking paid expenses when split among other household members/roommates
- Potential calculation errors caused by splitting expenses and accounting
- Costs are unpredictable
- Most people don't budget
- Financial knowledge is low

CURRENT PROCESS FLOW DIAGRAM



SOLUTION STATEMENT

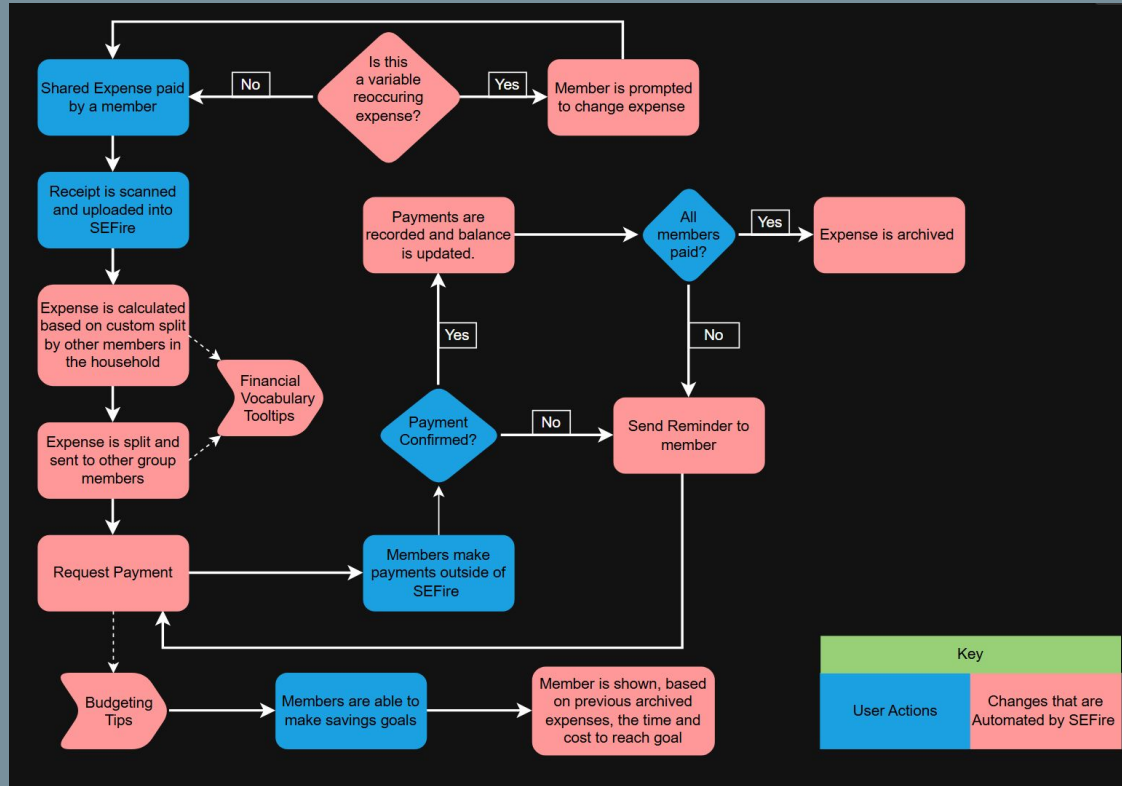
SEFire is a mobile friendly application that help roomates record shared expenses, calculate each members shared expenses and track outstanding balances. The application also informs its members of the importance of budgeting and explaining financial vocabulary using helpful tooltips.

SOLUTION PROCESS FLOW

Things that were added/changed -

- ❑ **Variable Expense Prompt:** Members are prompted on whether the expense made has changed (if the expense reoccurs)
- ❑ **Custom Expense Splitting:** Expense is calculated using custom splitting made before the expense is added into the system.
- ❑ **External Payments:** Members must pay outside of the application as the application does not handle money within the app itself
- ❑ **Group Payment Verification:** Since the app considers multiple members in a group, it verifies all members have paid their expense (if necessary) before archiving the expense as completely paid.
- ❑ **Budgeting Awareness:** Members are informed of the importance in having a budget and provide tips to help manage their expenses.
 - ❑ Members are also able to place a savings goal to work towards
- ❑ **Financial Vocabulary:** Members can view definitions of unfamiliar financial terms via tooltips.

SOLUTION PROCESS FLOW DIAGRAM



WHAT IT WILL DO

- ☐ Create a household group and invite members.
- ☐ Record bills with payer, total, and member sharing cost.
- ☐ Custom Splitting Expenses
- ☐ Custom Variable Expenses (Expected vs Actual)
 - ☐ Allows the input of “variable income” to determine expense expectation by next paycheck
- ☐ Itemizes receipts
- ☐ Show balances
- ☐ Records payments
- ☐ Send reminders for upcoming bills/subscriptions
- ☐ Reminder for individuals who has not paid for a specific expense

WHAT IT WILL NOT DO

- ☐ It wont pay your bills for you.
- ☐ Force someone to pay.
- ☐ No in app payments.
- ☐ Banking information will not be stored
- ☐ Will not facilitate transactions between users
- ☐ Won't manage nor teach people their personal finances
- ☐ Will not manage investments

COMPETITION MATRIX PART 1

Functionality	SEFire	Splitwise	Tricount	Settle Up	SplitCircle
Invite members / Create Groups	✓	✓	✓	✓	✓
Record shared bills	✓	✓	✓	✓	✓
Savings Goals via Budget tooltips	✓	✗	✗	✗	✗
Custom splitting expenses	✓	✓	✓	✓	✓
Variable Expense Prompt	✓	✗	✗	✗	✗
Show balances	✓	✓	✓	✓	✓

COMPETITION MATRIX PART 2

Functionality	SEFire	Splitwise	Tricount	Settle Up	SplitCircle
Record payments	✓	✓	✓	✓	✓
Send reminders	✓	✓	✓	Premium	✓
Notifications for those who have not paid	✓	✓	✓	Premium	✓
Financial Vocab/Tooltips	✓	✗	✗	✗	✗
Scan Receipts (OCR)	✓	Premium	✗	✗	Premium

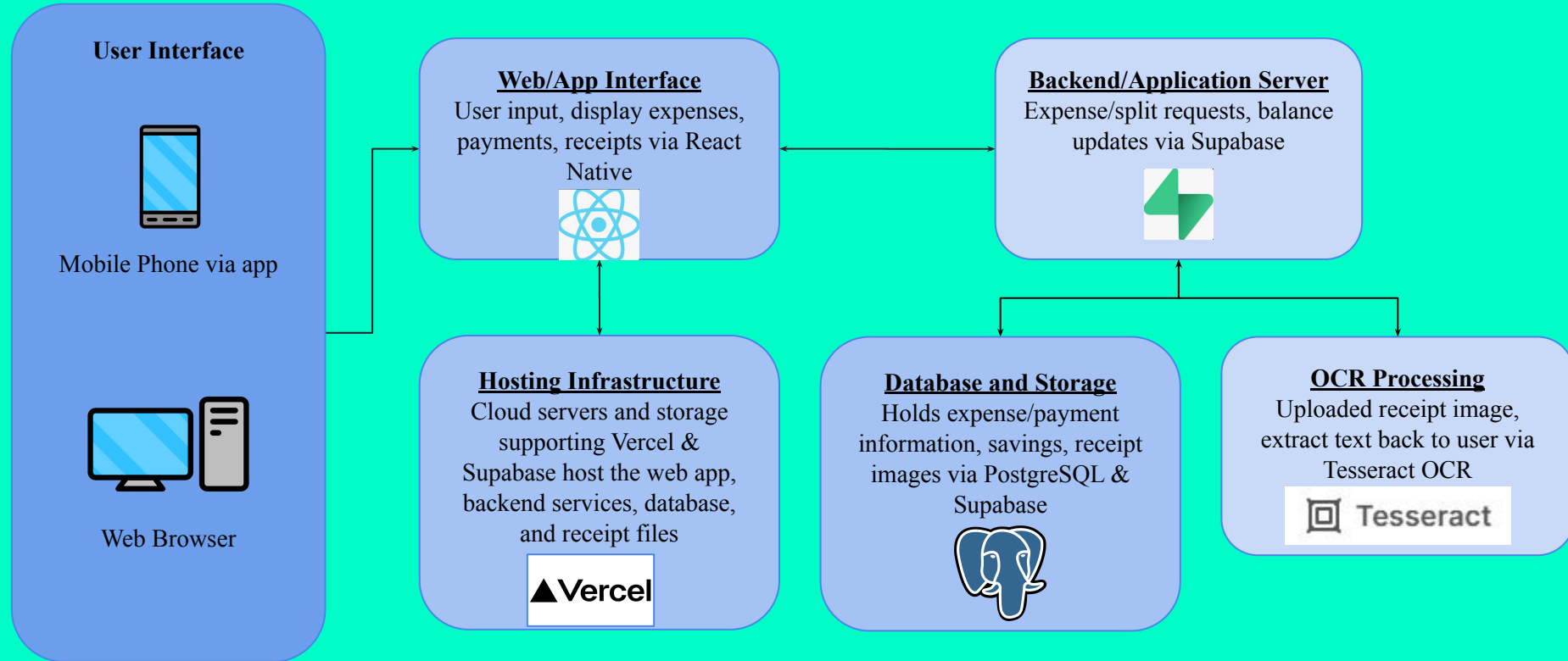
DEVELOPMENT TOOLS

- ❑ **Integrated Development Environment (IDE)** - VSCode
- ❑ **Version Control** - Git through GitHub.
- ❑ **Continuous Integration (CI) and Continuous Deployment (CD)** - GitHub Actions & Workflows
- ❑ **Frontend/UI** - React Native
- ❑ **Backend** - Supabase
- ❑ **Database** - PostgreSQL
- ❑ **Selected Language (Backend)** - SQL (PostgreSQL and Supabase), Python (Supabase), and JavaScript (for React Native)
- ❑ **Selected Language (Frontend)** - A combination of HTML, CSS, and JavaScript (webpage)
- ❑ **Testing Framework(s)** - PyTest for Python and Jest for JavaScript
- ❑ **Documentation Tool** - dbForge for PostgreSQL, pydoc for Python, and JSDoc for JavaScript

MAJOR FUNCTIONAL COMPONENTS

- ❑ **User Interface:** Mobile phone application and web browser, users can enter expenses, check balances, and track savings.
- ❑ **Web/App Interface:** React Native, renders SEFire's interface across mobile app and web instead of separate applications.
- ❑ **Backend:** Supabase handles user accounts, application logic for expenses, splitting costs, and updating balances.
- ❑ **Database:** PostgreSQL stores household membership, expenses, payment records.
- ❑ **OCR Processing:** Tesseract OCR extracts details from uploaded receipts, extracted information can be edited and saved.
- ❑ **Hosting Infrastructure:** Cloud servers and storage supporting Vercel and Supabase. Hosts the web application, backend services, database, and receipt files.

MAJOR FUNCTIONAL COMPONENTS DIAGRAM



RISKS

- **Customer and End User:** Low User Adoption
 - Users may prefer existing methods or competing expense-management apps. **Impact:** 4 | **Probability:** 3
 - Mitigation: Advertisement of the product including features and benefits. **Impact:** 3 | **Probability:** 2
- **Technical:** OCR Inaccuracy/ Integration Issues
 - Receipt data may be extracted incorrectly, and software components may encounter integration issues. **Impact:** 4 | **Probability:** 4
 - Mitigation: Users review and edit extracted information before saving. **Impact:** 3 | **Probability:** 2
- **Security:** Exposure of User Financial Data
 - Unauthorized access could expose users' financial or household information. **Impact:** 5 | **Probability:** 3
 - Mitigation: Authentication, access controls, and secure cloud storage. **Impact:** 5 | **Probability:** 1
- **Legal:** Privacy and Data Handling Requirements
 - User information and uploaded receipts must be stored and handled appropriately. **Impact:** 5 | **Probability:** 3
 - Mitigation: User agreements and terms and conditions agreements. **Impact:** 4 | **Probability:** 1
- **Financial:** Hosting & Service Costs
 - Growth in users could increase database, hosting, and third-party service costs. **Impact:** 3 | **Probability:** 3
 - Mitigation: Scalable hosting and usage monitoring. **Impact:** 2 | **Probability:** 2

Risk Table
1 - Negligible
2 - Low
3 - Medium
4 - High
5 - Extreme

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APPENDIX

REAL WORLD VS PROTOTYPE TABLE

Features & Functionality	RWP	Prototype (Planned)	Prototype (Actual)